

Amplifying individual differences in neural representation ?

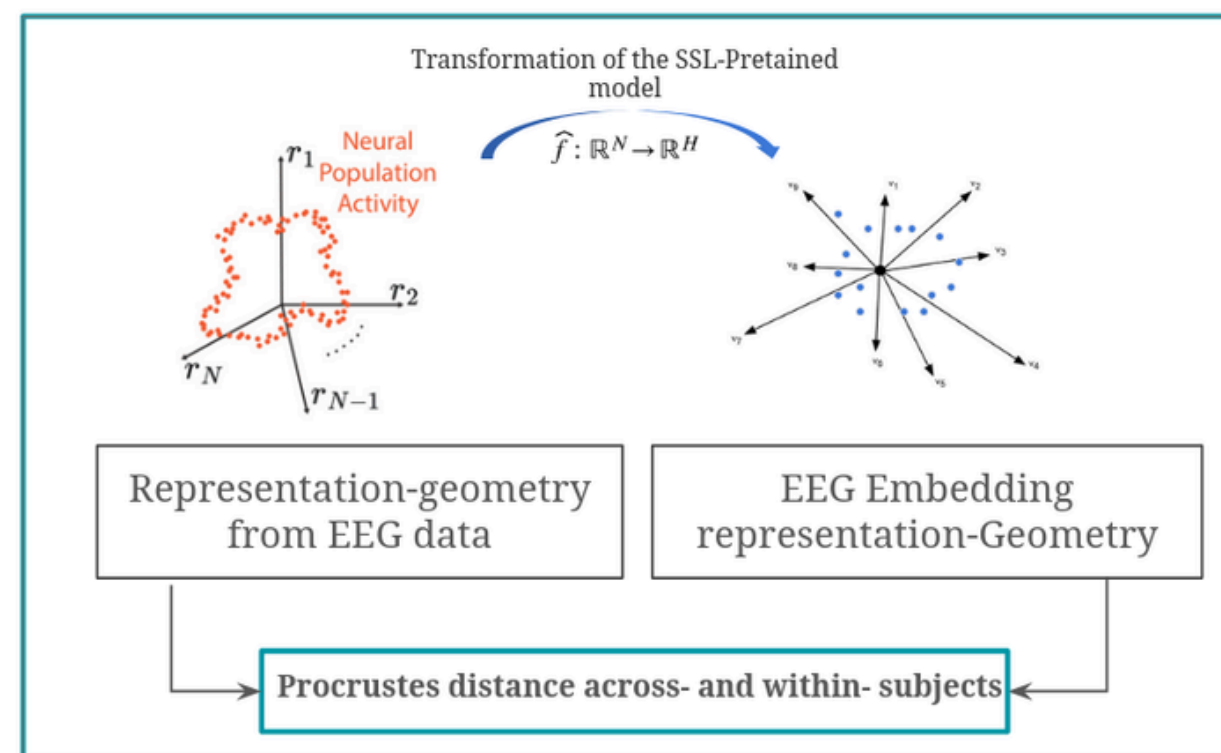
Self-supervised EEG Pretraining does it.

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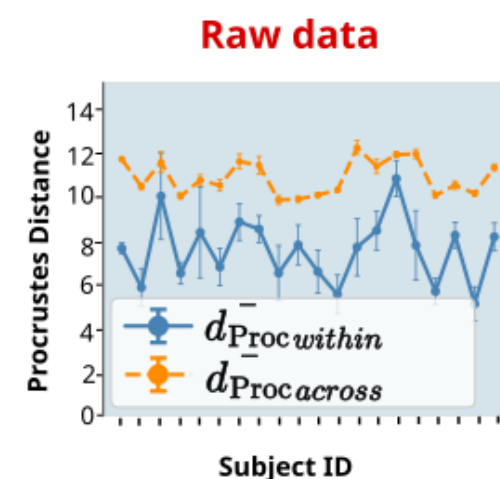


With neural data, **self-supervised learning** (SSL) can be leveraged to build **generalized representations** from massive, heterogeneous datasets. To go beyond well-known behavioral decoding metrics, we wonder : how **subject-specific idiosyncrasies** are affected by such large-scale pretraining ?

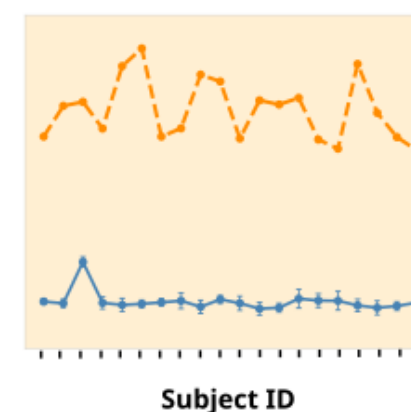
EEG : Pretraining **amplifies idiosyncrasies** and decodability



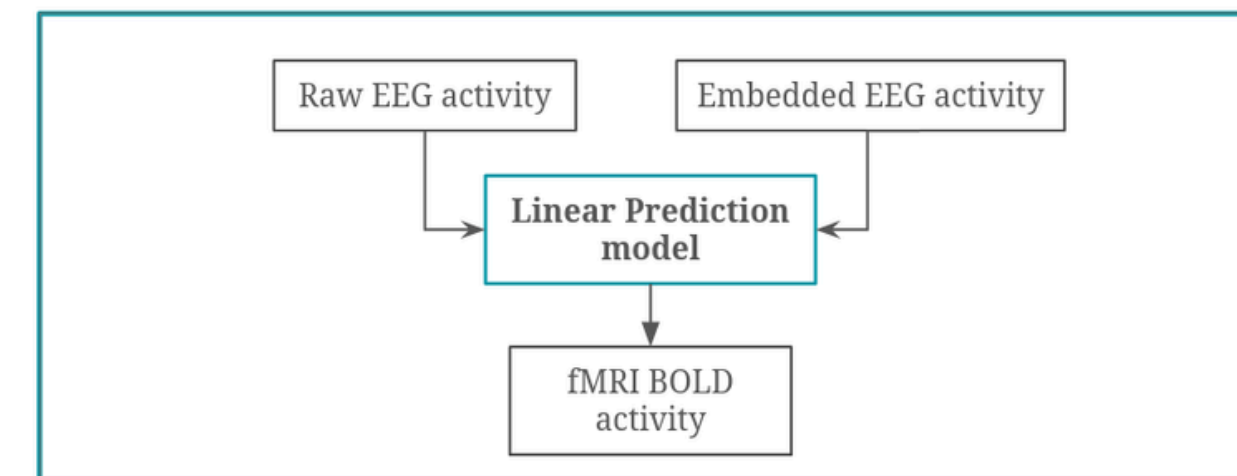
RESULTS 1



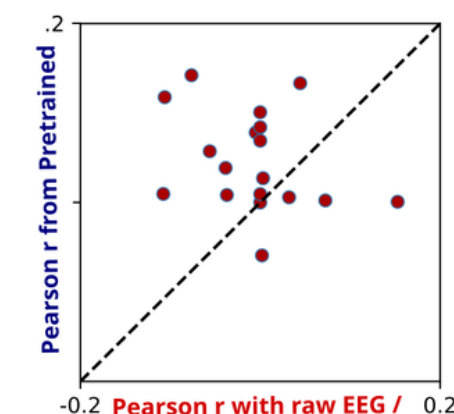
NeuroGPT Pretrained



EEG and fMRI : Pretraining enhances **cross-modal alignment**



RESULTS 2



← Predictive accuracy of BOLD activity measured with Pearson R (each scatter represents one subject)

CONTRIBUTION

- Analyzing the **representational geometry** of a foundation model trained with SSL
- Uncovering that SSL **amplifies individual differences** and improves both subject **identity** and **behavioral** decoding
- Showing that SSL improve **fMRI prediction from EEG** by carrying subject-specific **cross-modality information**
- ... and showing this is not merely a matter of dimensionality

THE DATA

Simultaneous recordings of EEG and fMRI on 18 subjects

